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Theme: AI for Climate Change

Climate Resilience Decision Intelligence Platform

An AI decision-support layer that turns fragmented climate data into actionable decisions that build climate resilience.

Team Climate Bandits

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AGENDA

Where we're headed

01

The Problem

Data-rich, decision-poor

02

HIDDEN TRADE OFFS

Why solving one problem creates another

03

Our Idea

A decision-intelligence platform

04

How It Works

Scenario simulation & optimization

05

Architecture

Data to recommendation, end to end

06

Nugen AI Layer

Reasoning over trade-offs

07

Impact & Scale

From one region to many

We are data-rich, but decision-poor

Governments and industries already collect enormous volumes of environmental data - weather, air and water quality, land use, infrastructure, energy, waste. The bottleneck isn't data collection anymore.

The real question isn't “what is happening?”

It's: given everything that's happening and how it's all connected — what should a decision-maker actually do next?

The gap isn't information. It's decision intelligence.

Abundant

environmental datasets already exist across agencies

Interconnected

risks : a fix in one system shifts pressure to another

Missing

A decision layer that connects the data to action.

Fixing one risk can quietly worsen another

A single-issue fix can ripple through the wider system. Example : a common flood response



The lesson isn't “don't build drainage.” It's that any single intervention needs to be evaluated against the whole interconnected system — water, heat, biodiversity and cost together, not in isolation.

Monitoring isn't deciding

EXISTING CLIMATE-TECH

Problem-specific systems

- **Flood systems —** predict / manage floods only
- **Air-quality platforms —** monitor pollution levels
- **Water systems —** track supply and stress
- **Energy platforms —** optimize consumption
- **Carbon trackers —** measure emissions

Each is useful. But they rarely answer the cross-system question: "What should we do?"

OUR APPROACH

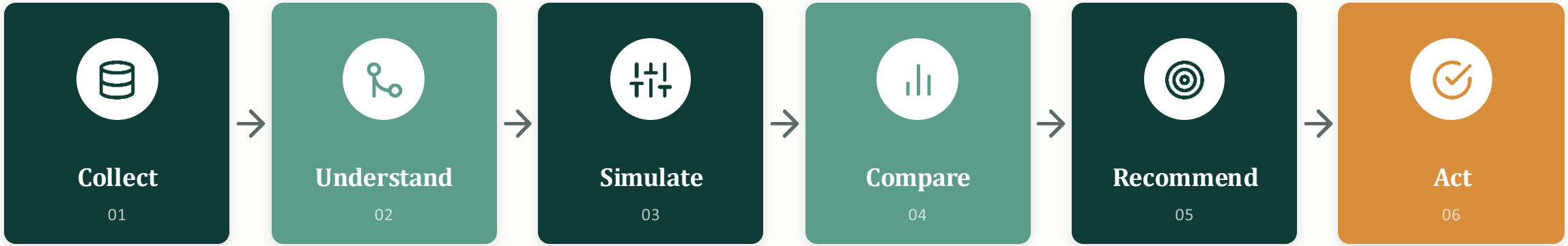
A cross-system decision layer

- **Connects across systems —** flood, water, heat, land use and more
- **Simulates interventions —** before money is committed
- **Optimizes trade offs —** resilience, cost, environment impact jointly
- **Explains the "why" —** reasoning, not just a dashboard
- **Keeps humans in control —** the system recommends , the officials decide

Environmental intelligence, converted into an actionable decision.

One platform, one continuous decision loop

An AI-powered Climate Resilience Decision Intelligence Platform that combines environmental data, simulates intervention scenarios, and recommends the most sustainable combination of actions by weighing resilience, environmental impact, cost and long-term consequences together.



We don't just predict environmental problems — we help decision-makers choose the best intervention before those problems become crises.

Two distinct layers, working together

DATA & MODEL LAYER

Quantitative estimates

- Cleans and fuses rainfall, water, heat, land-use and infrastructure data
- Learns relationships between environmental variables
- Simulates each intervention scenario numerically
- Multi-objective optimizer scores options on resilience, cost, environmental impact and feasibility

AI DECISION LAYER

Reasoning and trade-offs

- Reads scenario outputs from the model layers.
- Weighs competing objectives against stated priorities and budget constraints.
- Explains why one combination outperforms the rest.
- Presents options and trade-offs — the official makes the final call

A city has ₹50 crore and a recurring flood risk

Officials test interventions before committing budget. The system scores each on the outcomes that matter - not just flood reduction.

Option	Flood risk	Groundwater	Urban heat	Cost
A. Bigger drainage	↓↓↓	—	—	₹48 Cr
B. Wetland restoration	↓↓	↑↑	↓	₹22 Cr
C. Permeable surfaces	↓	↑	↓	₹15 Cr
D. Wetland + permeable (combined)	↓↓↓	↑↑↑	↓↓	₹35 Cr

Recommended: Option D. Near-equal flood protection to full drainage expansion at ₹35 Cr - while also improving groundwater recharge and reducing urban heat

From raw environmental data to a recommended action

Data Sources	Rainfall, AQ/WQ sensors, satellite/GIS, land use, historical records
↓	
Data & GIS Layer	Cleaning, spatial mapping, storage (PostGIS)
↓	
Environmental Intelligence	Risk/trend analysis, anomaly detection, environmental relationships
↓	
Scenario Engine	“What if we do X?” — single & combined interventions
↓	
Multi-Objective Optimizer	Resilience · carbon · water · heat. biodiversity · cost.
↓	
AI Decision Layer	Recommendation, reasoning, trade-offs (Nugen AI)
↓	
Interactive Dashboard	Risk map, scenario compare, cost/benefit, recommendation

KEY TECHNOLOGIES

Frontend

Next.js / React, Tailwind, Mapbox, Recharts

Backend

Python, FastAPI, REST APIs

Data

PostgreSQL + PostGIS

AI / ML

Pandas, NumPy, Scikit-learn, optimization, Nugen AI

Geospatial

GeoPandas, open source GIS & climate datasets

Not “AI predicts climate change.” This.

Relationship modelling

Represents environmental factors as a connected graph, so the system understands that changing one variable moves several others.

Scenario simulation

Turns an intervention idea into a testable “what-if” alone or combined with others before resources are committed.

Multi-objective optimization

Compares options on resilience, water, heat, biodiversity, cost and feasibility together, not one metric at a time.

Trade-off reasoning

Explains why a combination wins and the consequences, not just a ranked score.

Human-in-the-loop

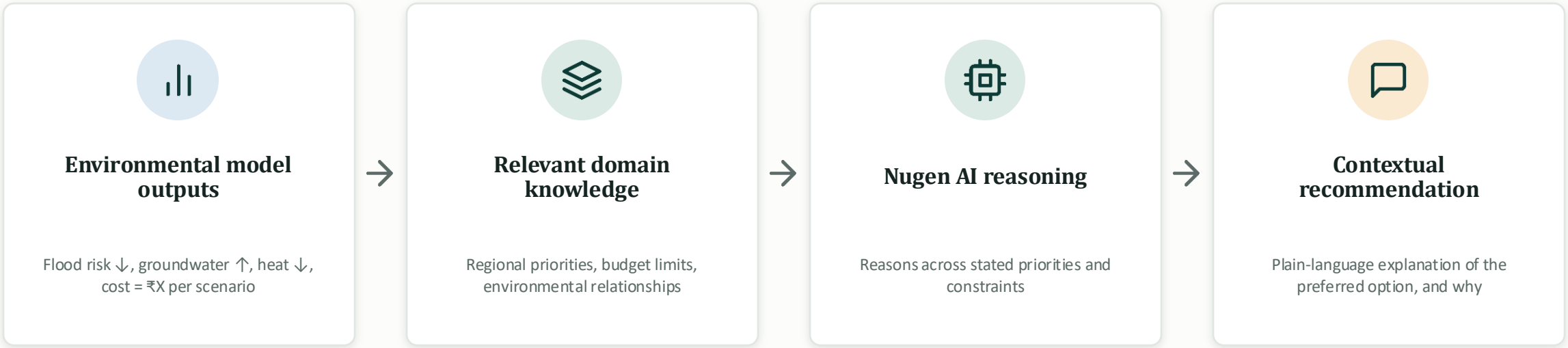
Surfaces options and consequences; the decision-maker chooses. The system never says “do this” on its own.

Reusable across risks

The same relationship graph and optimizer extend to heat, air quality, waste and energy without rebuilding system for each problem.

Nugen reasons over results — it doesn't replace them

Our own models compute the numbers. Nugen AI turns those numbers, plus domain context, into a plain-language recommendation an official can act on. **Nugen AI acts as our domain-aligned decision agent.**



Boundary: Nugen explains and contextualizes results our environmental and optimization models already computed — it does not perform the underlying simulation or scoring.

One good decision, several compounding benefits

Environmental

- Better water & land management
- Biodiversity protection
- Lower-carbon infrastructure choices
- Greater long-term climate resilience

Economic

- Better allocation of limited budgets
- Reduced cost from preventable damage
- Environmental benefit per ₹ spent
- Fewer solutions that create new problems

Operational

- Faster, evidence-backed decisions
- Reactive response → preventive action
- Multiple datasets in one framework
- Complex trade-offs made understandable

***Primary users** : Municipal authorities · Government planners · Industries · Environmental agencies*

***Ultimate beneficiaries** : Safer, healthier communities and more resilient ecosystems*

One reusable engine, expanding in every direction

01

Proof of Concept

1 region + 1 interconnected system — flooding, water security, urban heat

→

02

City Expansion

Expand to more neighbourhoods; add air quality, energy, waste, biodiversity, infrastructure

→

03

Multiple Cities

Same platform ingests each city's GIS, climate & infrastructure data

→

04

Larger Systems

River basins → industrial regions → states → countries

What stays constant: the core pipeline — data → risk understanding → scenario simulation → optimization → recommendation. New problems mainly need new datasets and intervention rules, not a new platform.

MVP Deliverables: Interactive climate decision web platform
Risk map · Scenario simulator · Multi-objective comparison · AI recommendation · Nugen-powered explanation

From monitoring climate risks to optimizing decisions



01 — Beyond prediction
We move from "what will happen?" to "what should we do?"

02 — Beyond single metrics
We optimize resilience, environment and cost together.

03 — Built to scale
One decision engine can extend across risks, regions and cities.

SUMMARY:
Our platform combines environmental data, AI/ML, scenario simulation and multi-objective optimization into one scalable decision layer - starting with one region, built to scale toward city-scale climate-resilient planning.

> ***"Don't just predict the next climate crisis. Decide what to do before it happens."***